

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo

Improving Efficiency

Selected Topics

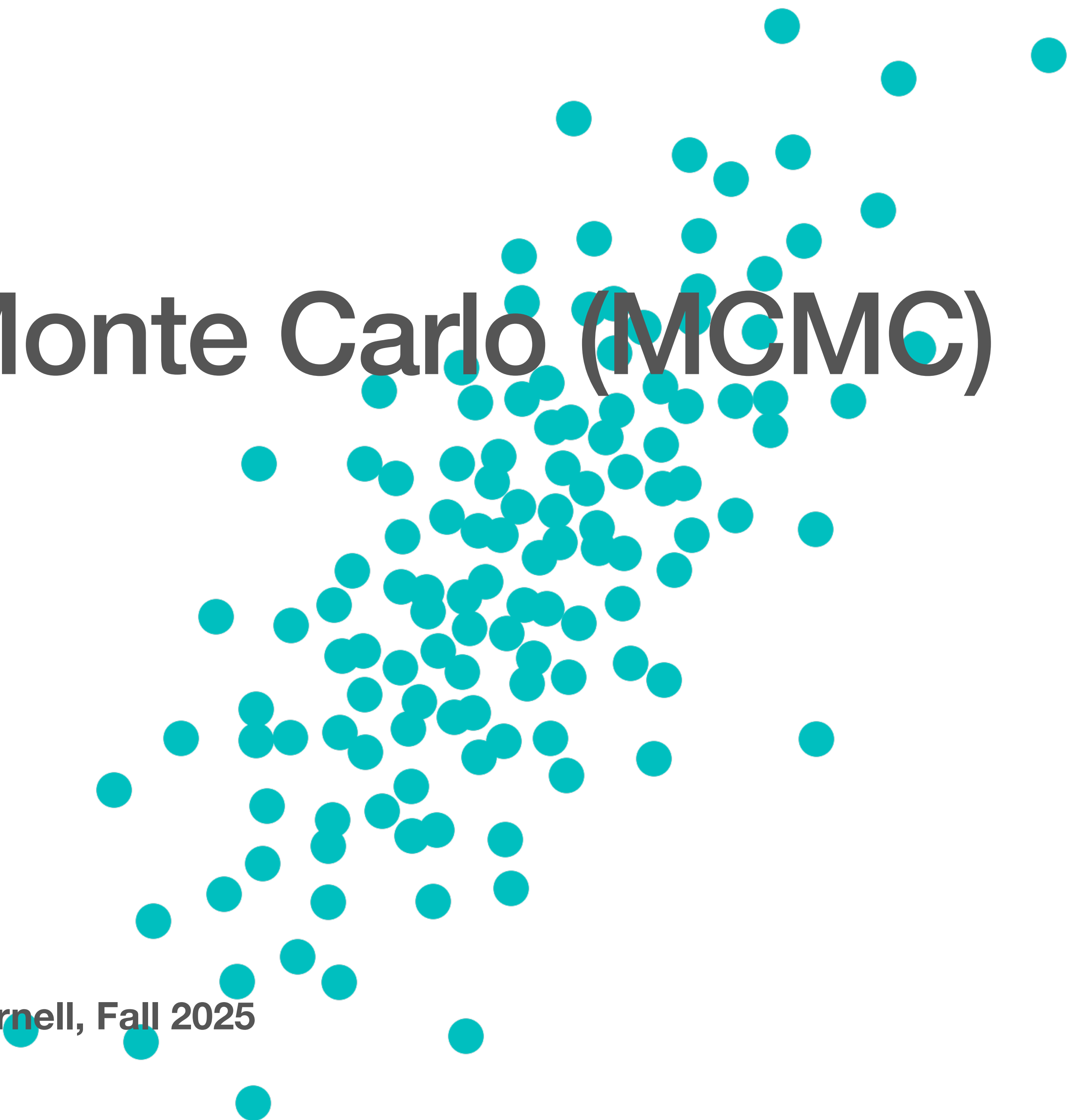
Git website and repository
Canvas

Fred Hickernell, Fall 2025

Updated 2025 September 5



Markov Chain Monte Carlo (MCMC)

A decorative graphic consisting of numerous teal-colored dots of varying sizes, scattered across the right side of the slide. The dots are more densely packed in the center-right area and become sparser towards the top and bottom right corners.

Owen, Chapters 11– 12

Assignment 3 due ??

Project Selection due Oct 1



How do we generate samples when the distribution is complicated?



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 - An unnormalized target density



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 - May reject a lot of samples



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- **Inverse cumulative distribution** function applied to uniform samples works for simple cases
- **Correlated multivariate Gaussian** distributions are possible through affine transformations of univariate Gaussians
- **Acceptance-rejection** sampling
 - Propose from a simple density that you know bounds
 - An unnormalized target density
 - May reject a lot of samples
- **Markov chain Monte Carlo** (MCMC) moves samples toward areas of higher target density



Metropolis–Hastings algorithm

Suppose that

- You really **want to** sample X with known (unnormalized) **target** density q
- You have a **proposal** density $q_{\text{new|old}}$ to select the next point starting from where you are

Given $X[0]$

For $i = 0$ to $n-1$

Generate $Z \sim q_{\text{new|old}}(\cdot | X_i)$, $U \sim \mathcal{U}[0,1]$

If $U \leq \min\left(1, \frac{q(Z)q_{\text{new|old}}(X_i | Z)}{q(X_i)q_{\text{new|old}}(Z | X_i)}\right) = \min\left(1, \frac{\text{joint density of first } Z \text{ then } X_i}{\text{joint density of first } X_i \text{ then } Z}\right)$

$X[i+1] \leftarrow Z$

Else

$X[i+1] \leftarrow X[i]$

Note that you **always accept** either the new or the old



Metropolis algorithm

If $q_{\text{new}|\text{old}}(x | z) = q_{\text{new}|\text{old}}(z | x)$, then this can be simplified

Metropolis sampling proceeds as follows

Given $X[0]$

For $i = 0$ to $n-1$

Generate $Z \sim q_{\text{new}|\text{old}}(\cdot | X_i)$, $U \sim \mathcal{U}[0,1]$

If $U \leq \min\left(1, \frac{q(Z)}{q(X_i)}\right)$

$X[i+1] \leftarrow Z$

Else

$X[i+1] \leftarrow X[i]$



Advantages of and Challenges in MCMC

- Samples from complicated, unnormalized densities
- Easy to implement
- Proposal density and starting point need tuning to ensure that
 - Explore all the sample space and not miss out
 - Exploit sampling in regions where the density is large



How do we measure quality of MCMC

Short answer: We cannot easily

Longer answer: Given a symmetric, positive definite kernel, $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, the **discrepancy**, sometimes called the maximum mean discrepancy between two empirical distributions is defined by

$$D^2(\{\mathbf{x}_i\}_{i=0}^{n-1}, \{\mathbf{z}_i\}_{i=0}^{m-1}) = \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{x}_i, \mathbf{x}_j) - \frac{2}{nm} \sum_{i,j=0}^{n-1, m-1} K(\mathbf{x}_i, \mathbf{z}_j) + \frac{1}{m^2} \sum_{i,j=0}^{m-1} K(\mathbf{z}_i, \mathbf{z}_j)$$

Requires $\mathcal{O}(\max(n, m)^2)$ operations